CHAPTER 2

Elementary Particles

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This chapter gives the reader an insight in the world of "sub-electron" particles. It is in no way exhaustive but is intended to tease the interest of the reader into studying more literature dedicated to this specific subject. The basics of nuclear physics is of great importance to understand the mechanism governing the physics inside a nucleus.

The electron, proton and neutron are well known elementary particles. However, high energy nuclear collisions, such as those that occur in the atmosphere when it is hit by very high energy particles from space, are known to produce a wealth of other particles which once used to be called elementary particles. Such particles can now also be formed using large particle accelerators on earth, e.g. the one in CERN.

2.1. ELEMENTARY PARTICLES

The group of elementary particles began to be considerably expanded about 1947 when physicists discovered the first of the so-called "strange" particles in cloud chamber pictures of cosmic rays, see Figure 2.1. These new elementary particles were called strange because they lived almost a million million times[1] longer than scientists had any reason to expect at that time. The population of elementary particles has literally exploded since then, as physicists have built larger and larger particle accelerators, by which it is possible to impart sufficient kinetic energy to protons so that interaction with nuclides transform a large fraction of the kinetic energy into matter. The present limit is in the multi TeV range (Ch. 16). In 2011 it is thus possible to produce particles with a mass of up to ~3500 proton masses, and hundreds of new "strange" particles have been observed. Figure 2.2 is a typical picture of a reaction observed in a liquid hydrogen bubble chamber at an accelerator center.

[1] million million times = 10^{12} times

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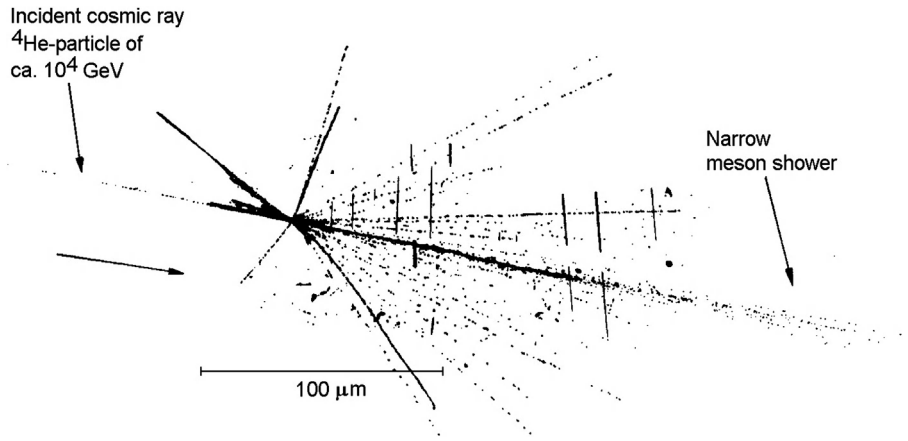


Figure 2.1 Secondary particles produced by a 10^4 GeV helium atom in a photographic emulsion.

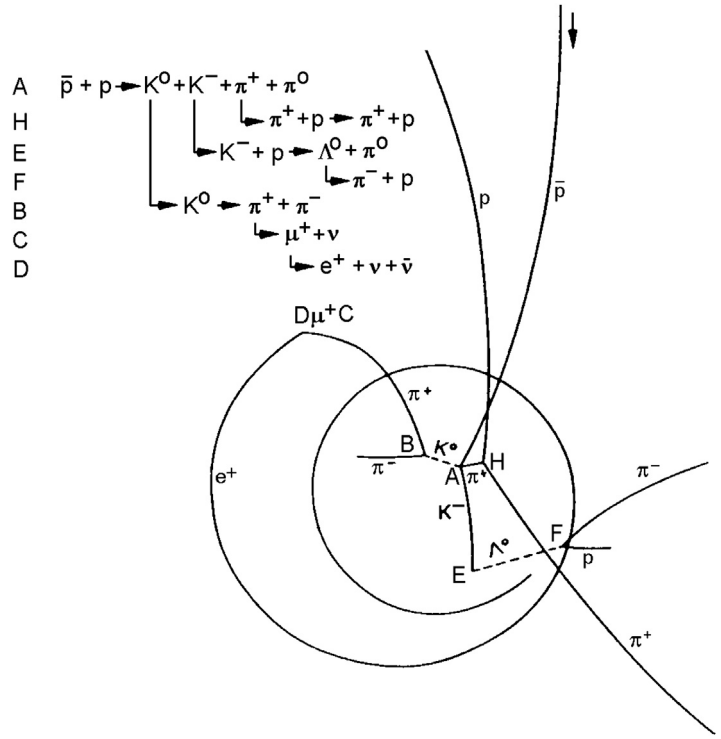


Figure 2.2 The reaction products of an annihilated antiproton as seen in the CERN liquid hydrogen bubble chamber. (*Annual report 1961*, CERN).

This has created a scientific area called *elementary particle physics*. It is quite different from nuclear physics, which is concerned with composite nuclei only. A principal objective of elementary particle physics has been to group the particles together according to their properties to obtain a meaningful pattern, which would describe all particles as parts of some few fundamental building blocks of nature. One step in this direction is to study how the elementary particles interact with each other, i.e. what kind of forces are involved.

Scientists have long doubted that all the particles produced with masses between the electron and the proton (loosely referred to as *mesons*, i.e. "intermediate"), and with masses greater than the proton (referred to as *baryons*, "heavy") really are "elementary". It was proposed that they have a substructure or constitute excited states of each other. Are they waves or particles since they serve as carriers of force? At this point it is important to understand what is meant by "particle" in nuclear physics.

2.2. FORCES OF NATURE

Considering what an immense and incredible diverse assembly the universe is — from the cosmos to man and microbes — it is remarkable that scientists have been able to discover with certainty only four basic forces, which govern the attraction and repulsion of all physical objects of nature. Let us consider these *forces of nature* in a qualitative way from the weakest to the strongest, see Table 2.1.

The first and weakest force of nature is that of *gravity*. The gravitational force, F_g, beween two masses, m_1 and m_2, at a distance r is given by eqn (2.1), where **G** is the universal constant of gravity ($\mathbf{G} = (6{,}6742 \pm 0{,}0007) \times 10^{-11}$ N m^2 kg^{-2}).

$$F_g = \mathbf{G} \times m_1 \times m_2 / r^2 \tag{2.1}$$

F_g is the force that causes all objects to always attract one another and is responsible for the attraction of the planets to the sun in the solar system and of the solar system to the rest of the galaxy. It is also the force that holds us to the earth. It seems paradoxical that

Table 2.1 Forces of nature in order of strength and their exchange particles

Force	Relative strength	Range	Exchange particle	Spin	Rest mass
1. Gravitation	1	very long	graviton[†]	2	0
2. The weak force	10^{25}	very short	W^+, W^-	1	79.9 GeV
			Z^0	1	91.2 GeV
3. The electromagnetic force	10^{36}	long	photon	1	0
4. The nuclear force (the strong force)	10^{38}	10^{-14} m	gluon	1	0

[†]Not yet detected.